# Wiqaya: Wildfire Susceptibility Mapping and Alerts for the Forests of Northern Morocco

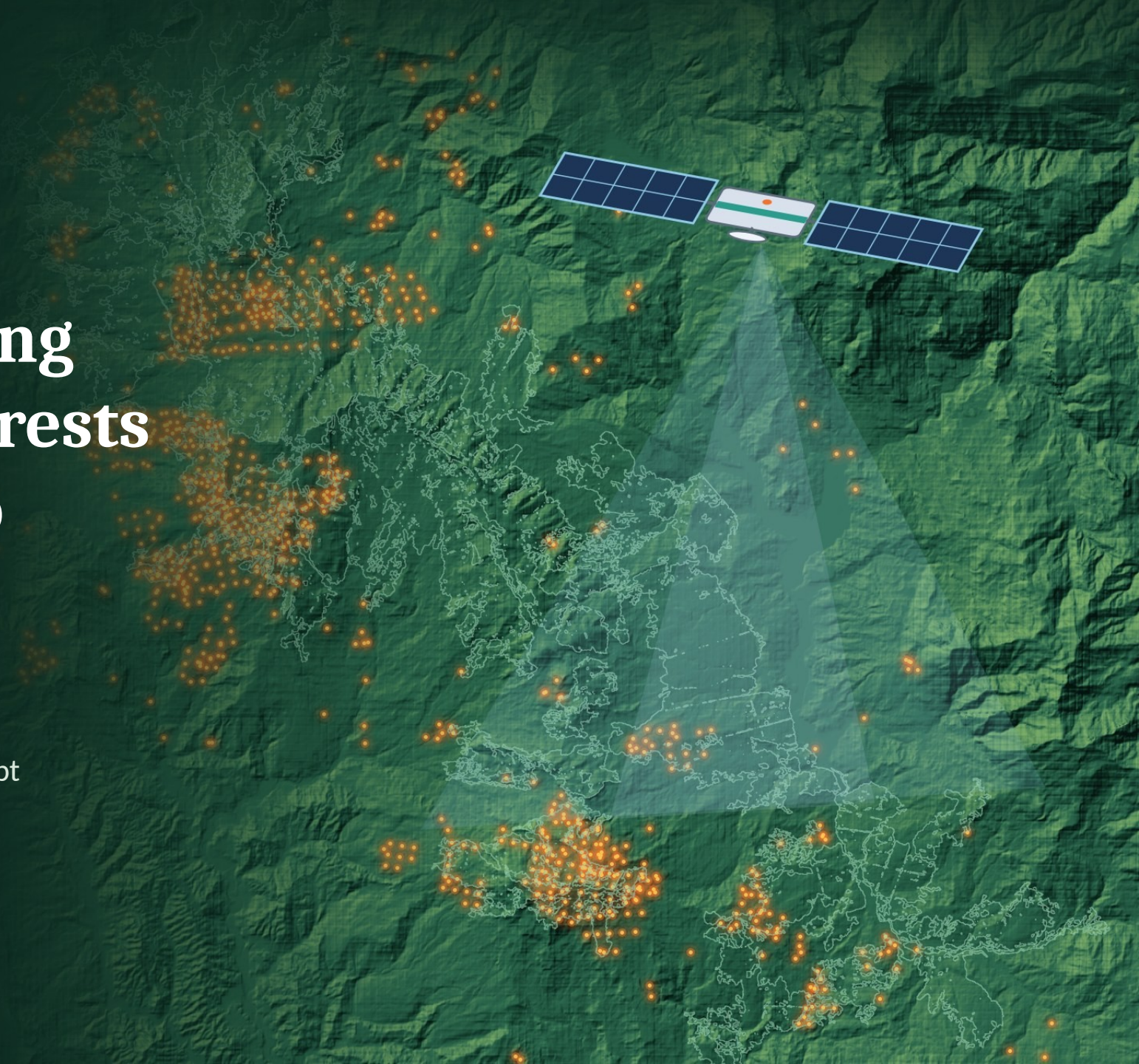
**Wiame Kane · Sanae Belfrouh · Yassine Azmou · Mohammed Yassine Sof · Hajar Elouatiq**

Team Wiqaya — Morocco

Arab Youth Space Hackathon 2026 · Proof of Concept

*Theme: Climate & Disaster Resilience*

October 2026 · github.com/SanaeBelfrouh/wiqayagit



# The western Rif burns almost every summer, and warnings arrive after ignition

- **Recent large fires:** Bouhachem 2022 (> 7,500 ha), Larache 2022 (> 10,500 ha), Derdara 2021 and Bouhachem 2020 (~1,100 ha each), two fires in August 2025
- **Drivers:** hot, dry chergui winds, steep slopes, villages (douars) inside the forest
- **Gap:** danger bulletins are province-scale and satellite detections come after ignition; there is no daily, cell-level view of risk
- **Why satellite data:** NASA FIRMS gives the only consistent 20-year fire record of these remote mountains (MODIS since 2005, VIIRS 375 m since 2017)

**10,600**
people living inside the forest

**264,000**
people within 10 km of it

**445**
FIRMS detections inside the forest since 2005

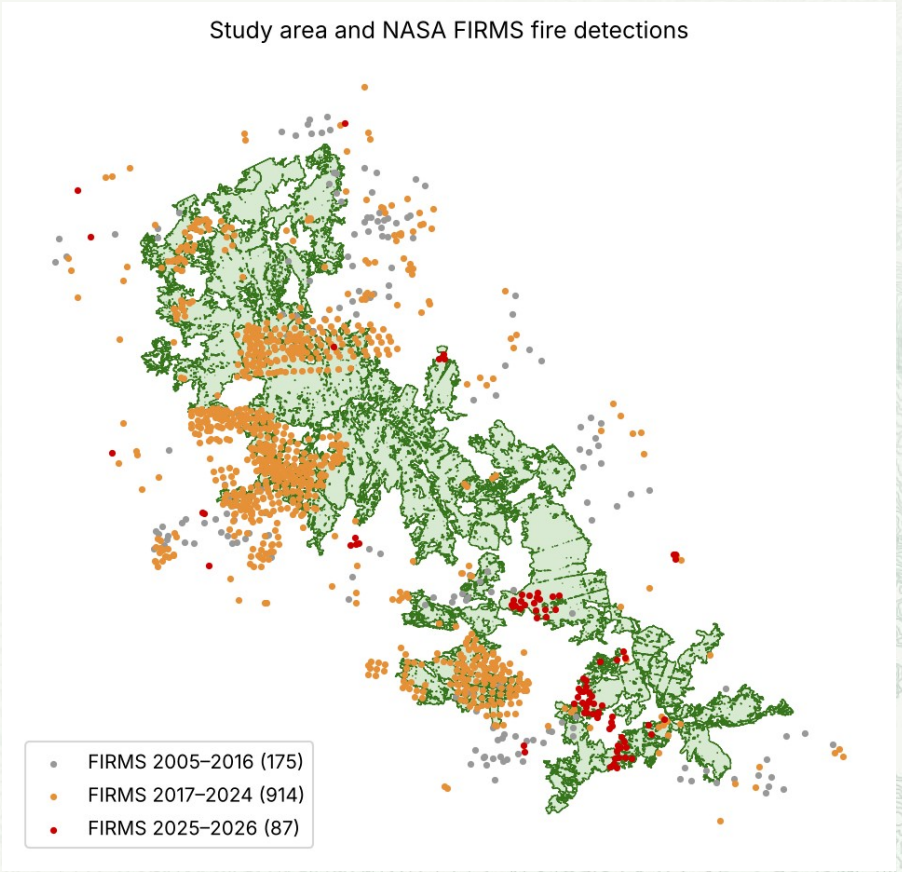



*Fig. 1. Study area (25,900 ha) and NASA FIRMS detections by period. Population: WorldPop 2017.*

# Business use case: where to patrol today, and whether to warn residents

| | |
|---|---|
| **End users** | Forest managers (national forestry agency, ANEF), Civil Protection, provinces and communes; residents of the villages in and around the forest |
| **Decision supported** | Each summer morning: where to send patrols and first-intervention teams, and whether the day is dangerous enough to warn residents and restrict burning |
| **Current practice** | Province-scale danger bulletins, field knowledge, and fire detections (FIRMS, EFFIS) that arrive after ignition; no alerts in Arabic and French |
| **Contribution of Wiqaya** | 250 m risk map with alert zones and a per-cell explanation, daily weather danger index, 20-year fire history, SMS / email alerts in three languages, open GeoJSON API |
| **Business model** | Subscription per protected area for institutions; free citizen alerts funded by these contracts or CSR partners; open data, so no imagery cost |

# Data: open Earth-observation and ancillary datasets

| Dataset | Product and provider | Period | Use |
|---|---|---|---|
| **Active fires** | NASA FIRMS: MODIS C6.1 and VIIRS S-NPP 375 m | 2005–2026 (1,176 points) | Labels, fire history |
| **Terrain** | FABDEM V1-2, 30 m (Univ. of Bristol; Copernicus GLO-30) | 2023 release | Elevation, slope, aspect |
| **Population** | WorldPop mar_ppp_2017 (CC BY 4.0) | 2017 | Exposure per cell |
| **Weather** | Open-Meteo: ERA5 archive and forecast (CC BY 4.0) | 2017–today | Weather danger index |
| **Optical imagery** * | Sentinel-2 L2A, Landsat C2 L2 (Planetary Computer) | 2005–2025 | NDVI, NDMI, NBR, dNBR |
| **Land cover, OSM** * | 2021 classification; OpenStreetMap 2022 | 2021–2022 | Fuel, human pressure |

* *Research notebook only.* **Hyperspectral data is not used in this proof of concept;** EnMAP or Arab Satellite 813 scenes are planned to map live fuel moisture and vegetation species.

# Approach: from satellite detections to alert zones

**1. Inputs**
FIRMS 2005–2026, DEM, WorldPop, forest boundaries

**2. Grid and features**
6,071 cells of 250 m; elevation, slope, aspect, population

**3. Fire history**
FIRMS within 375 m; years since last fire (no leakage)

**4. Model selection**
Logistic regression vs random forest, leave-one-year-out 2017–2024

**5. Fuel rule**
Risk 0 if burned < 1 year ago; × 0.5 if 1–5 years

**6. Blind test**
Retrain on 2017–2024; score the 2025 fires

**7. Risk map**
5 KMeans classes; alert = riskiest 20 % of cells

**8. Weather danger**
Max T, VPD, wind, min RH ranked vs 2017–2024 summers

*Reproducible: notebooks/02_main_analysis.ipynb runs end to end in about 5 minutes from the committed sample inputs (fixed seeds, relative paths).*

# Risk classes on 1 August 2025: the 2025 fires fell outside the alert zones

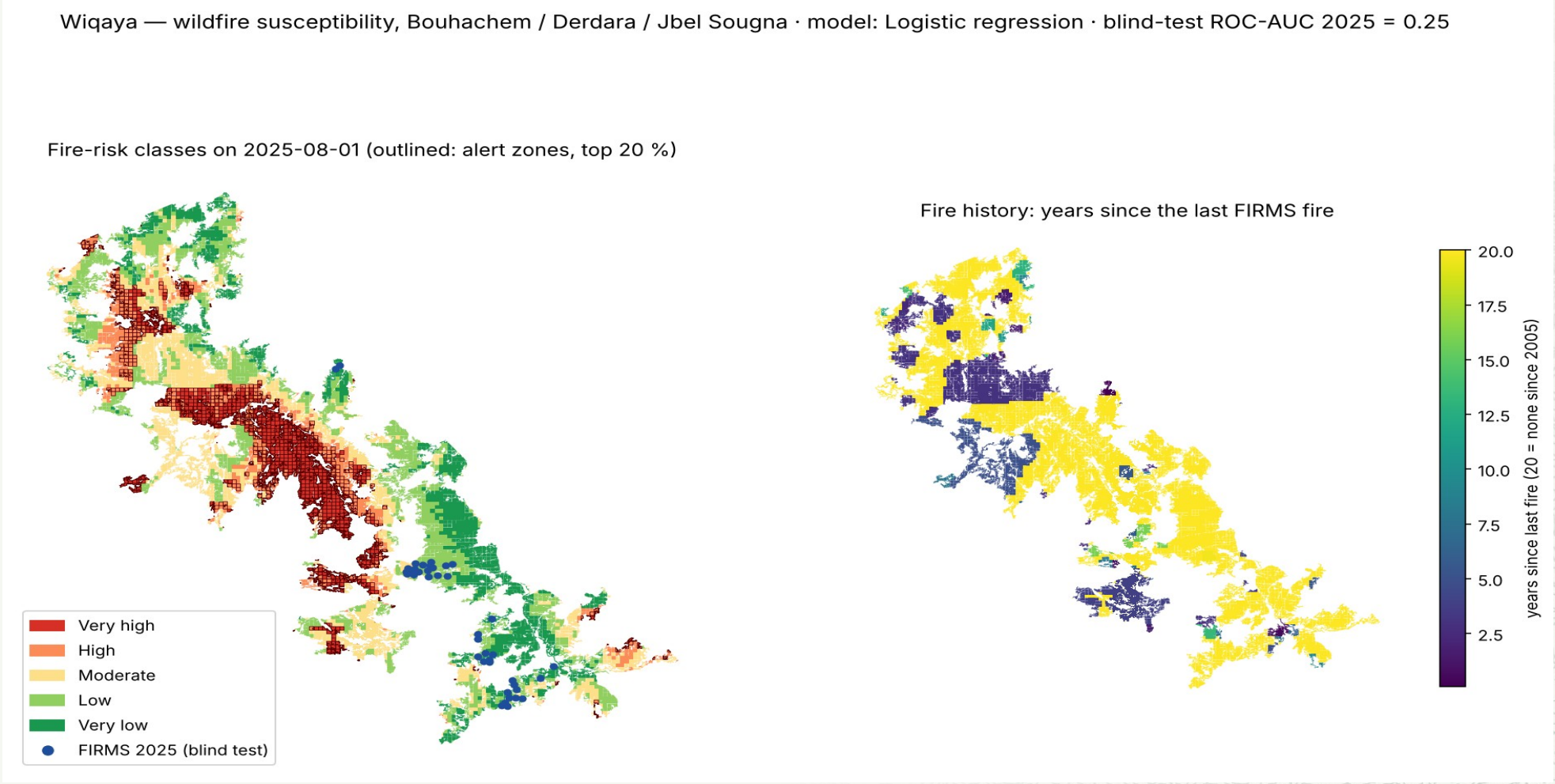



*Fig. 2. Left: five risk classes; outlined cells are alert zones (riskiest 20 %); blue points are the 2025 FIRMS fires (blind test). Right: years since the last FIRMS fire (20 = none since 2005). Notebook output results/example_output.png.*

# Population, terrain and fire history drive the selected model




*Fig. 3. Absolute standardised coefficients of the logistic regression.*

- **Selected model:** L2 logistic regression (C = 0.5, balanced classes)
- **Strongest predictors:** population, elevation, east-facing aspect, then fire history (years since the last fire, burned in the last 5 years)
- **Weak predictor:** slope
- **Missing drivers:** vegetation dryness and human access, available only in the research notebook

# The same model is delivered to users through a web platform

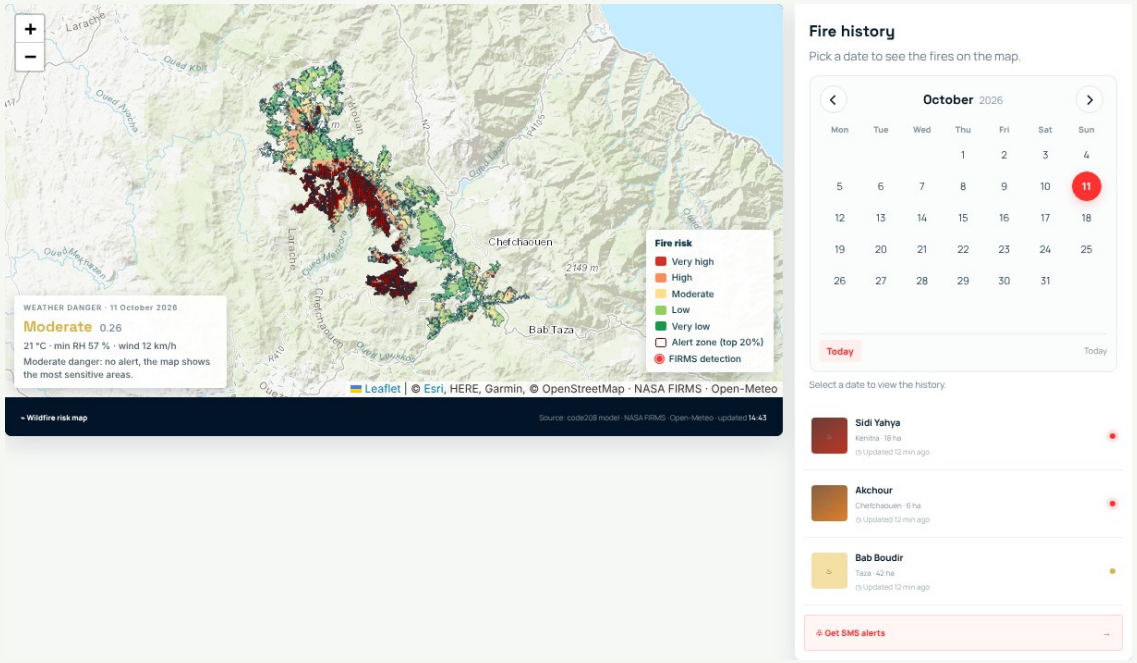



*Fig. 4. Daily risk map with the weather danger of the day and the five-class legend.*

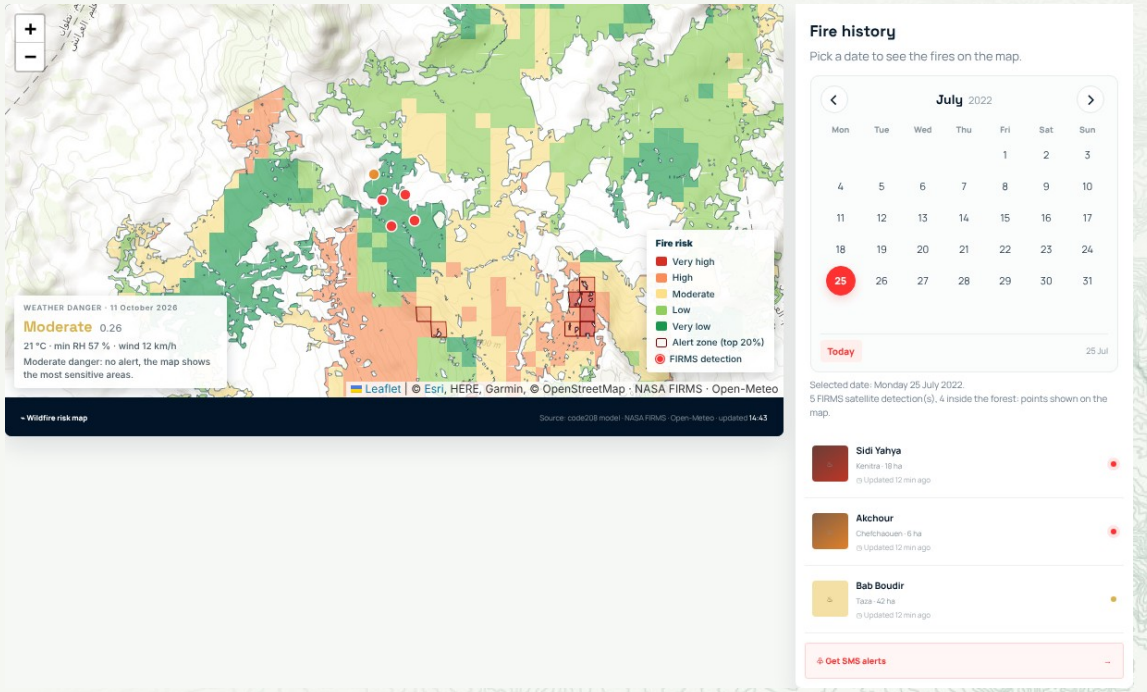



*Fig. 5. Fire history: 25 July 2022, start of the Bouhachem fire; FIRMS shows 5 detections, 4 inside the forest.*

# Access for residents: Arabic, French and English, on any phone

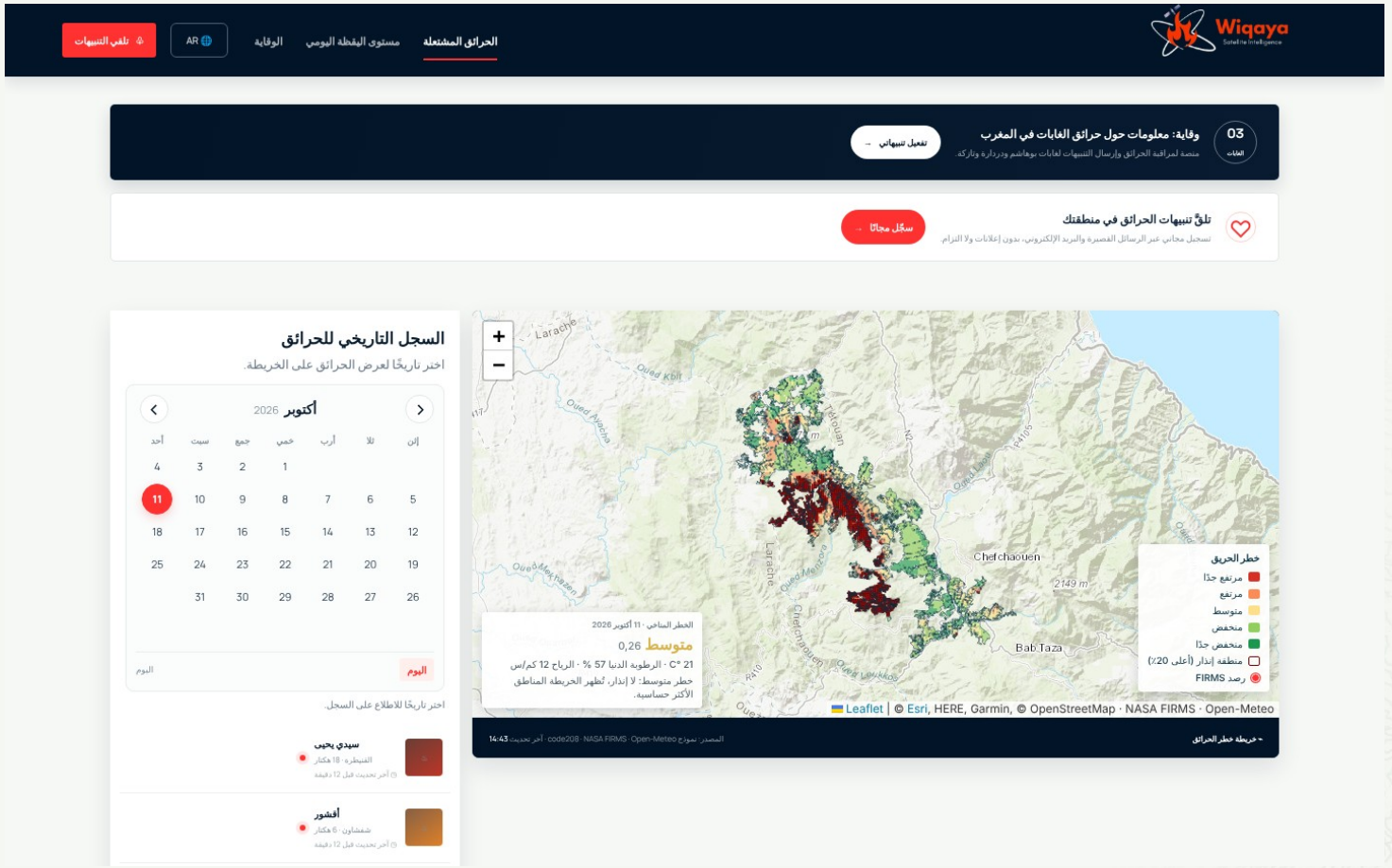



*Fig. 6. Right-to-left Arabic interface.*

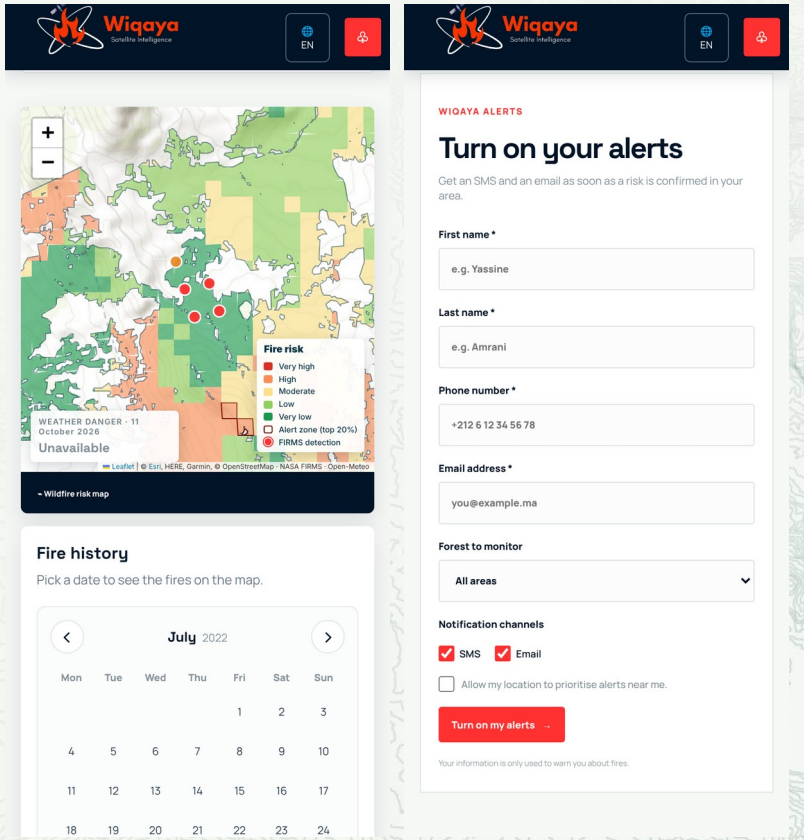



*Fig. 7. Fire history and SMS / email alert sign-up on a phone.*

# Validation: informative on large-fire years, but the 2025 blind test fails

| Test | Result |
|---|---|
| Leave-one-year-out 2017–2024, logistic regression (selected) | Mean ROC-AUC per year 0.62; global 0.65; PR lift 1.64 |
| Leave-one-year-out 2017–2024, random forest | Mean ROC-AUC per year 0.56; global 0.60 |
| ROC-AUC per held-out year | 2018: 0.99 · 2019: 0.88 · 2020: 0.80 · 2021: 0.68 · 2023: 0.23 · 2024: 0.06 |
| **Blind test, 2025 fires (39 cells)** | **ROC-AUC 0.25; no fire cell in the alert zones** |

- **Works** on the large fire years 2018–2021
- **Fails** on 2023–2025: the 2025 fires started in cells rated low risk
- **Cause:** predictors describe exposure, terrain and history, not vegetation dryness or human access
- **Few positives:** 345 fire cell-years; per-year scores are unstable
- **No leakage:** history strictly before each date; 2025 never used for training or selection

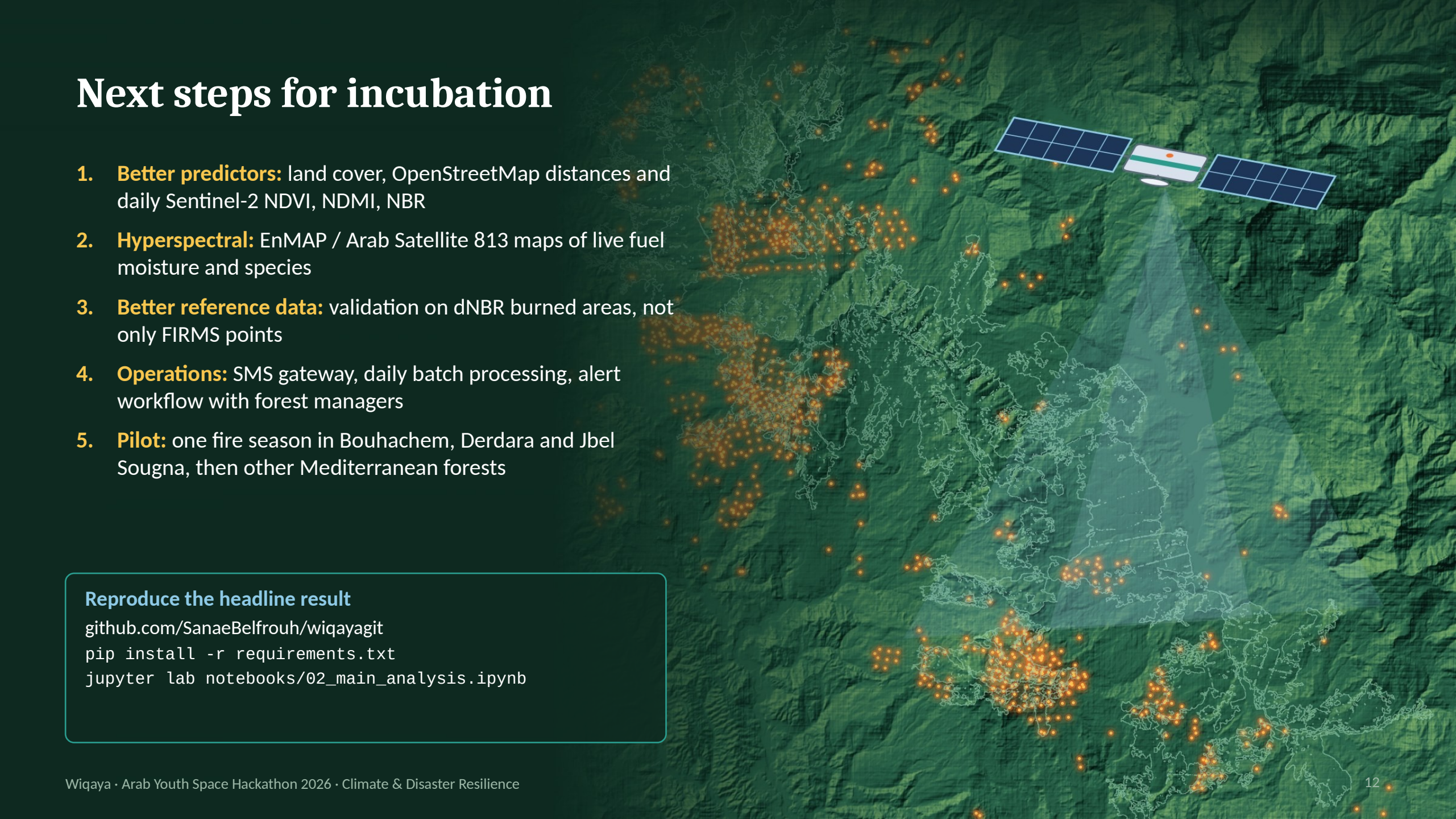
# Expected impact and strategic alignment

| Goal | Contribution |
|---|---|
| **SDG 13 Climate action (13.1)** | Resilience to heatwave- and wind-driven fire seasons |
| **SDG 15 Life on land (15.1, 15.2)** | Patrols and firebreak work targeted at the riskiest cells |
| **SDG 11 Sustainable communities (11.5)** | Early warning for ~10,600 people living inside the forest |

- **Morocco:** supports the strategy Forêts du Maroc 2020–2030, ANEF and Civil Protection
- **Arab region:** open data only; replicable in Mediterranean forests of Algeria, Tunisia, Lebanon and Syria
- **Inclusion:** Arabic-first interface; SMS for residents without smartphones

# Next steps for incubation

1. **Better predictors:** land cover, OpenStreetMap distances and daily Sentinel-2 NDVI, NDMI, NBR
2. **Hyperspectral:** EnMAP / Arab Satellite 813 maps of live fuel moisture and species
3. **Better reference data:** validation on dNBR burned areas, not only FIRMS points
4. **Operations:** SMS gateway, daily batch processing, alert workflow with forest managers
5. **Pilot:** one fire season in Bouhachem, Derdara and Jbel Sougna, then other Mediterranean forests



**Reproduce the headline result**

github.com/SanaeBelfrouh/wiqayagit

```
pip install -r requirements.txt
jupyter lab notebooks/02_main_analysis.ipynb
```

# References and data sources


Assali, F. et al. (2016). Forest-fire risk mapping in the Chefchaouen-Ouazzane region, Morocco (topoclimate index).

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Software: GeoPandas, Rasterio, rasterstats, scikit-learn, FastAPI, Leaflet. Code released under the MIT licence.